

SC1008 C/C++ Programming
NTU BSc Computer Science — Comprehensive Textbook (0–100)

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Chapter 1

C Basics: Your First Conversation with the Machine

“C is not a high-level language to do low-level things — it is a low-level language with high-level syntax.” Before objects, templates, or smart pointers, we must speak the machine’s own dialect. This chapter builds it from zero.

From zero: no prior programming assumed. We start with what a program *is* (instructions acting on memory), then build types, variables, I/O, and control from first principles with pictures before formalism. Later chapters depend only on what is built here.

Learning Outcomes

After this chapter you will be able to:

- (L1) describe the compile–run pipeline and the role of `main()`;
- (L2) declare variables with correct C types and explain their memory cost;
- (L3) read and write data with `printf/scanf`;
- (L4) write branching (`if/switch`) and loops (`for/while`);
- (L5) trace execution and predict output of short C programs.

1.1 What Is a C Program?

A computer is a *stored-program* machine: a long array of memory cells, each holding a byte (8 bits), plus a CPU that fetches, decodes and executes instructions one by one. A C program is a recipe written in text that we *compile* (translate) into those machine instructions.

Every C program starts at `main()`:

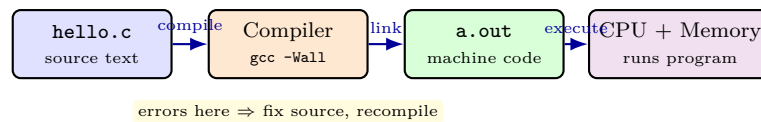


Figure 1.1: The C pipeline: source $\rightarrow$ compiler $\rightarrow$ executable $\rightarrow$ running process. Errors at compile time are cheap; errors at run time are bugs.

```

1 #include <stdio.h>           // header: declarations for printf/scanf
2 int main(void) {             // entry point, returns exit status
3     printf("Hello, NTU!\n");
4     return 0;                 // 0 = success
5 }
  
```

`#include` pastes in declarations; `int main` is the door the operating system knocks on.

1.2 Types and Variables: Boxes in Memory

A *variable* is a named box occupying contiguous bytes in memory. Its *type* says how many bytes and how to interpret the bits.

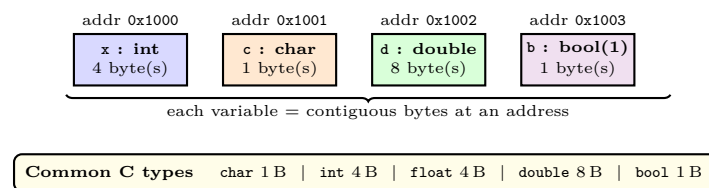


Figure 1.2: Variables as typed byte-boxes in memory. The type dictates size and interpretation; the address is where the box lives.

```

1 int    age = 19;              // 4 bytes, signed -2^31 .. 2^31-1
2 char   grade = 'A';          // 1 byte, single quotes
3 double gpa = 4.75;           // 8 bytes, floating point
4 _Bool  done = 0;              // 0 false, nonzero true (#include <stdbool.h> gives bool)
5 const int MAX = 100;          // const: cannot be reassigned
  
```

C is *statically typed*: the type is fixed at declaration. Initialise always — an uninitialised `int` holds garbage (whatever bits were there before).

Integer promotion and truncation matter: assigning 300 to a `char` keeps only the low 8 bits. Use `sizeof(int)` to query size portably.

1.3 Input and Output

`printf` formats output; `scanf` parses input. Both use *format specifiers* that must match the variable type.

```

1 #include <stdio.h>
2 int main(void){
3     int age; double gpa;
4     printf("Enter age and GPA: ");
  
```

```

5   if (scanf("%d %lf", &age, &gpa) != 2){ // & = address-of (Ch.2)
6       printf("Bad input!\n"); return 1;
7   }
8   printf("Age %d, GPA %.2f\n", age, gpa); // %.2f = 2 decimal places
9   return 0;
10  }

```

Common specifiers: %d int, %c char, %f float, %lf double, %s string (char array). Always check `scanf`'s return value: it counts successful conversions.

Buffering note: `printf` may not appear until newline `\n` or `fflush(stdout)`.

1.4 Control Flow: Branching and Loops

Programs need decisions and repetition. C provides both with minimal syntax.

1.4.1 Branching

```

1   int score = 82;
2   if (score >= 80)    printf("A\n");
3   else if (score >= 70) printf("B\n");
4   else                printf("C\n");
5
6   char op = '+';
7   switch(op){         // switch: jump table on integer/char
8       case '+': printf("add\n"); break;
9       case '-': printf("sub\n"); break;
10      default:  printf("other\n");
11  }

```

`break` exits the `switch`; without it execution *falls through*. Braces `{}` group multiple statements.

1.4.2 Loops

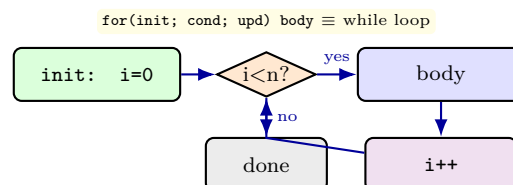


Figure 1.3: Flow of a for loop: initialise $\rightarrow$ test $\rightarrow$ body $\rightarrow$ update $\rightarrow$ test ... until the condition fails.

```

1   for (int i = 0; i < 5; i++)    printf("%d ", i); // 0 1 2 3 4
2
3   int k = 5;
4   while (k > 0){ printf("%d ", k); k--; } // 5 4 3 2 1
5
6   int n;
7   do { printf("Enter positive n: "); scanf("%d",&n); } while(n <= 0);

```

for is for counted iteration; **while** for condition-driven; **do-while** runs at least once. **break** exits the innermost loop; **continue** skips to next iteration.

1.5 Functions and Scope

A *function* is a named sub-recipe with parameters and a return value. It enables decomposition and reuse.

```

1  int max2(int a, int b){ return (a > b) ? a : b; } // ternary operator
2  int main(void){
3      int x = max2(3, 7);    // call: arguments copied into parameters
4      printf("%d\n", x);    // 7
5  }
```

C passes arguments *by value*: the function receives copies. To let a function modify the caller's variable, pass its address (pointer) — the subject of Ch. 2. Variables have *block scope*: visible only inside their `{}`.

1.6 Operators, Expressions, and Precedence

An *expression* combines values and operators to produce a new value. C evaluates using precedence (higher binds tighter) and associativity.

Prec.	Operator	Associativity
1	() [] -> .	left
2	! ~ ++ - + - * & (unary)	right
3	* / %	left
4	+ -	left
5	< <= > >=	left
6	== !=	left
7	&&	left
8		left
9	?:	right
10	= += -=	right

```

1  int a=5,b=2,c=3;
2  int r = a + b * c;    // 11, not 21: * before +
3  int s = (a+b)*c;      // 21: parentheses override
4  int t = a > b && b < c; // 1 (true): relational before logical
5  // Pitfall: == vs = (comparison vs assignment)
6  if(a == 5) printf("yes\n"); // correct
7  // if(a = 5) always true (assigns 5, yields 5 which is truthy) -- bug!
```

Integer division truncates towards zero: `7/2==3`, `7%2==1`. Mixing signed/unsigned promotes to unsigned — a frequent bug source. When in doubt, parenthesise.

1.7 Worked Example: Grading Program

We combine all ideas into a small program that reads scores and prints a grade, demonstrating I/O, branching, loops, and functions.

```

1  #include <stdio.h>
2  char grade_of(int score){
3      if(score>=80) return 'A';
4      if(score>=70) return 'B';
5      if(score>=50) return 'C';
6      return 'F';
7  }
8  int main(void){
9      int n; printf("How many scores? ");
10     if(scanf("%d",&n)!=1 || n<=0) return 1;
11     for(int i=0;i<n;i++){
12         int s; printf("Score %d: ", i+1);
13         scanf("%d",&s);
14         printf(" -> Grade %c\n", grade_of(s));
15     }
16     return 0;
17 }
```

Trace with input 2 85 62: $n=2$, loop $i=0$ reads 85 $\rightarrow$ A, $i=1$ reads 62 $\rightarrow$ C. Each function call gets its own frame (Ch. 2).

1.8 Chapter Notes

Kernighan & Ritchie *The C Programming Language* (2nd ed.) remains the canonical reference. `gcc -Wall -Wextra -std=c11` enables all warnings — treat warnings as errors early.

1.9 Exercises

- E1.1 **Trace.** What does this print? `int a=5,b=2; printf("%d %d %d",a/b, a%b, (double)a/b);` Explain each expression.
- E1.2 **scanf pitfalls.** Why does `scanf("%d", age)` (without `&`) often segfault? What does checking `scanf`'s return value protect against?
- E1.3 **Loop rewrite.** Rewrite `for(int i=0;i<n;i++) sum+=i;` as a `while` loop and as a `do-while` loop. When do the three forms behave differently?
- E1.4 **Fall-through bug.** Write a `switch` on `char grade` that maps `'A'/'B'/'C'/'D'/'F'` to messages, correctly using `break`. Show a version where a missing `break` causes a wrong message.
- E1.5 **Function scope.** Write `int factorial(int n)` iteratively and recursively. Call both from `main` and explain why the recursive version needs a base case; what happens without it?

Chapter 2

Pointers and Memory: Addresses as Data

“Pointers are the C programmer’s superpower and the beginner’s nightmare.” A pointer is just an address held in a variable. Master this one idea and C’s memory model becomes transparent.

From zero: we build on Ch. 1 boxes. Memory is a numbered array of bytes; a variable’s name hides its address. We reveal the address, store it, and draw every step. No prior systems knowledge assumed.

Learning Outcomes

After this chapter you will be able to:

- (L1) distinguish a variable’s value from its address (& vs. *);
- (L2) draw stack frames and explain call-by-value vs. call-by-reference;
- (L3) perform pointer arithmetic correctly for typed pointers;
- (L4) identify dangling, null, and wild pointers;
- (L5) explain the stack vs. heap distinction at a high level (detail in Ch. 4).

2.1 Addresses and the & Operator

Every byte in memory has an index — its *address* (like a house number). A variable `int x=42` occupies 4 bytes starting at some address, say `0x7ffc1000`. `&x` yields that address; `*p` follows a pointer to the box it points at.

```
1 int x = 42, y = 7;
2 int *p = &x;           // p holds address of x; type int* = pointer to int
3 printf("%p %d\n", (void*)p, *p); // 0x7ffc1000 42
4 *p = 100;               // writes through p: x is now 100
5 printf("x=%d\n", x);    // 100
6 int **pp = &p;          // pointer to pointer
```

`*p` is an *lvalue*: you can assign to it. `NULL` (i.e. `(void*)0`) means “points nowhere” — dereferencing it crashes.

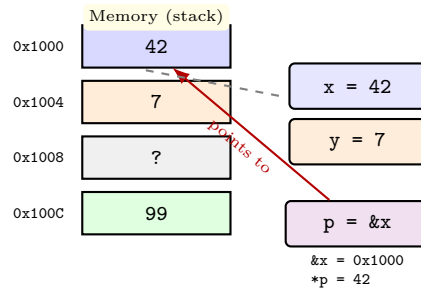


Figure 2.1: Memory as numbered boxes. `p` stores the address of `x`; dereferencing `*p` reads box `0x1000`.

2.2 The Stack: Frames on Call

Each function call pushes a *frame* (activation record) onto the call stack: parameters, local variables, return address.

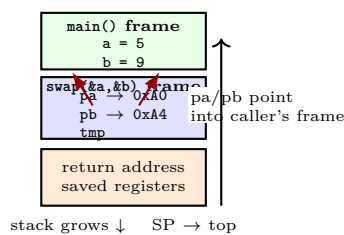


Figure 2.2: Stack frames: `main` calls `swap(&a, &b)`; the callee's pointers reach back into the caller's frame to modify `a, b`.

Call-by-value copies; to modify the caller, pass the address:

```

1 void swap(int *pa, int *pb){
2     int tmp = *pa; *pa = *pb; *pb = tmp;
3 }
4 int main(void){
5     int a=5, b=9; swap(&a, &b); // a=9 b=5
6 }

```

Without pointers, `swap(a, b)` would swap copies and the caller would see no change — a classic bug.

2.3 Pointer Arithmetic: Typed Steps

A pointer's type determines its stride. If `p` is `int*`, then `p+1` advances by `sizeof(int)` bytes (typically 4), not 1. The compiler scales for you.

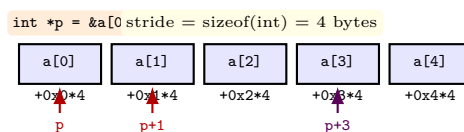


Figure 2.3: Typed pointer arithmetic: `p+1` jumps one `int` (4 bytes), not one byte. `a[i]` is sugar for `*(a+i)`.

```

1 int a[5]={10,20,30,40,50};
2 int *p = a;           // array decays to &a[0]
3 printf("%d\n", *(p+2)); // 30 == a[2]

```

```

4 | printf("%d\n", p[3]);    // 40 == *(p+3)
5 | char *cp = (char*)p;    // cp+1 advances 1 byte
6 | printf("%ld %ld\n", (long)(p+1)-(long)p, (long)(cp+1)-(long)cp); // 4 1

```

`void*` has no stride (cannot do arithmetic); cast first. Subtracting two `int*` gives element distance, not byte distance.

2.4 Dangerous Pointers

`NULL` dereference, *wild* (uninitialised), *dangling* (points to freed/reclaimed stack frame), and *out-of-bounds* pointers are undefined behaviour. Always initialise: `int *q=NULL`; and check before dereferencing. Never return the address of a local variable.

2.5 Const Pointers and Pointer-to-Const

C distinguishes three related types; the position of `const` matters.

```

1 | int x=5, y=9;
2 | const int *p1 = &x;    // pointer to const int: *p1 cannot be changed, p1 can
3 | int *const p2 = &x;    // const pointer to int: p2 cannot be changed, *p2 can
4 | const int *const p3 = &x; // both const
5 | // p1 = &y; ok      *p1=10; error
6 | // p2 = &y; error *p2=10; ok

```

Read declarations right-to-left: “`p1` is a pointer to a `const int`”. Use `const int*` for read-only views of data you do not own.

2.6 Arrays, Pointers, and Function Parameters

An array name decays to a pointer in most contexts, but `sizeof` and `&` are exceptions (Ch. 3 elaborates). When you pass an array to a function, you really pass a pointer — the callee cannot know the length.

```

1 | void print_arr(const int *a, int n){
2 |     for(int i=0;i<n;i++) printf("%d ", a[i]); // a[i] == *(a+i)
3 |     // a is really int*: sizeof(a)==8 on 64-bit, not 20!
4 | }
5 | int main(void){
6 |     int arr[5]={1,2,3,4,5};
7 |     printf("sizeof arr=%zu\n", sizeof(arr)); // 20
8 |     print_arr(arr, 5);                      // arr decays to &arr[0]
9 | }

```

Always pass the length explicitly. For 2-D arrays, all dimensions except the first must be known: `void f(int a[][3], int rows)`.

2.7 Worked Example: Pass-by-Reference via Pointers

C has only pass-by-value. To let a function return two values, pass output pointers:

```

1 #include <stdbool.h>
2 bool divmod(int a, int b, int *q, int *r){
3     if(b==0) return false;
4     *q = a/b; *r = a%b; return true;
5 }
6 int main(void){
7     int q,r;
8     if(divmod(17,5,&q,&r)) printf("17/5 = %d rem %d\n", q,r); // 3 rem 2
9 }

```

The caller lends the address; the callee writes through it. This pattern — output parameters via pointers — pervades C APIs (`scanf` itself uses it).

2.8 Pointer Safety Checklist

Before dereferencing any pointer, ask four questions:

1. Is it initialised? (wild if not)
2. Is it NULL? (null dereference crashes)
3. Does it still point to live storage? (dangling if freed/returned local)
4. Is the access in bounds? (overflow corrupts neighbours)

Adopt the habit: initialise to NULL, check before use, null after `free`, never return locals. Tools like AddressSanitizer (`-fsanitize=address`) turn silent UB into immediate aborts with stack traces — use them during development.

```

1 int *p = NULL;           // 1. initialise
2 p = malloc(sizeof(*p));
3 if(!p) return;           // 2. check allocation
4 *p = 42;                 // 3. live and in bounds
5 free(p); p = NULL;       // 4. free + null (prevents dangling)
6 // if(p) *p = 1;         // safe: null check guards

```

2.9 Chapter Notes

The stack/heap split is elaborated in Ch. 4. For deeper C memory, see Bryant & O'Hallaron *CS:APP* Ch. 3, 9.

2.10 Exercises

- E2.1 **Trace.** `int x=3; int *p=&x; int y=*p+1; *p=10;` What are `x`, `y`, `*p` after each line? Draw the boxes.
- E2.2 **Swap failure.** Explain why `void bad_swap(int a,int b){int t=a;a=b;b=t;}` does not swap the caller's variables. Fix it with pointers.
- E2.3 **Arithmetic.** On a machine with 4-byte `int`, if `p=0x1000` is `int*`, what are `p+1`, `p+5` numerically? What about `(char*)p+1`?
- E2.4 **Dangling.** Why is `int *f(){int x=5; return &x;}` dangerous? What might go wrong when the caller uses the returned pointer?
- E2.5 **Double pointer.** Write `void set_to_zero(int **pp)` that allocates no memory but makes `*pp = NULL` visible to the caller. Why does `void foo(int *p){p=NULL;}` not do the same?

Chapter 3

Arrays, Strings, and Structs: Building Bigger Boxes

“Bad programmers worry about the code. Good programmers worry about data structures.” — Linus Torvalds. We now aggregate bytes into arrays, text into strings, and records into structs.

From zero: arrays are just many boxes in a row. We start with contiguous storage, show why indexing is arithmetic, then build strings (char arrays + terminator) and structs (heterogeneous boxes) with pictures before code.

Learning Outcomes

After this chapter you will be able to:

- (L1) declare, initialise, and index arrays and explain bounds checking (or lack thereof);
- (L2) manipulate C strings safely with `<string.h>` functions;
- (L3) define and use `struct` types with correct member access;
- (L4) relate arrays to pointers and explain decay and `sizeof` pitfalls;
- (L5) design a small record type and its helper functions.

3.1 Arrays: Contiguous Homogeneous Storage

An array packs N elements of the same type contiguously. The name `a` is the address of element 0; `a[i]` means `*(a+i)`.

```
1  int a[5] = {11,22,33,44,55};    // initialise
2  int b[5] = {0};                // all zero
3  // int c[5] = {1,2};           // remainder zero-filled
4  for(int i=0;i<5;i++) printf("%d ", a[i]);
5  printf("\nsize=%zu bytes\n", sizeof(a)); // 20 = 5*4
```

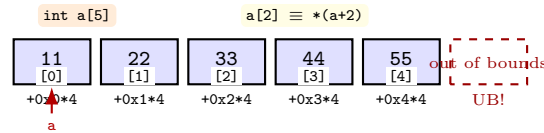


Figure 3.1: Array layout: 5 contiguous ints. Indexing is pointer arithmetic. Accessing beyond [4] is undefined behaviour (no bounds check in C).

```
6 // void f(int x[5]) -- inside f, sizeof(x) == sizeof(int*) !
```

C does *no* bounds checking. `a[5]` or `a[-1]` compiles but reads/writes arbitrary memory — a classic security hole. Multi-dimensional arrays are arrays of arrays: `int m[2][3]` is 6 ints row-major.

3.2 C Strings: Char Arrays with a Sentinel

A C string is a `char` array terminated by `'\0'` (value 0). The terminator marks the end; the array length may be larger.

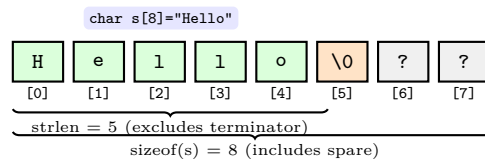


Figure 3.2: A C string "Hello" in an 8-byte buffer. `strlen` counts to `\0`; `sizeof` counts the whole array.

```
1 #include <string.h>
2 char s[16] = "Hello";           // 6 bytes used, 10 spare
3 printf("%zu %zu\n", strlen(s), sizeof(s)); // 5 16
4 strcpy(s, "Hi");                // copies including \0
5 strcat(s, "!");                 // appends: "Hi!"
6 // DANGER: strcpy with no bounds check -- use strncpy/strncpy or snprintf
7 snprintf(s, sizeof(s), "%s", "Hello, NTU!"); // safe, always terminates
8 for(int i=0; s[i]!='\0'; i++) putchar(s[i]);
```

String literals like "Hi" live in read-only memory; writing to them is UB. Always ensure a `\0` and enough space.

3.3 Structs: Heterogeneous Records

A `struct` groups different types into one compound type. Each *member* has its own offset.

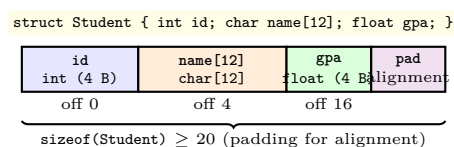


Figure 3.3: Struct layout: members placed in order with possible padding for alignment. `sizeof` includes padding.

```
1 struct Student{
```

```

2     int    id;
3     char   name[32];
4     float  gpa;
5 };
6 struct Student s = {123, "Asha", 4.8f};
7 printf("%s %.1f\n", s.name, s.gpa);
8 struct Student *ps = &s;
9 printf("%d\n", ps->id);           // -> for pointer to struct
10 // array of structs
11 struct Student cohort[50];
12 cohort[0].id = 1;

```

Access via `s.member` or `ps->member` (shorthand for `(*ps).member`). The compiler may insert padding between members for alignment; never assume `sizeof` equals sum of members. Use `typedef` to alias: `typedef struct Student Student;`

3.4 Arrays vs. Pointers — Decay

In most expressions, an array *decays* to a pointer to its first element. Exceptions: `sizeof(a)`, `&a`, and string literal initialisation. Hence function parameters declared as `int a[]` are really `int *a`; the callee cannot know the length — pass it explicitly: `void print(int *a, int n)`.

3.5 Searching and Sorting Arrays

With arrays we can implement linear search and bubble sort — the algorithmic core of SC1007, previewed here.

```

1 int linear_search(const int *a, int n, int key){
2     for(int i=0; i<n; i++) if(a[i]==key) return i;
3     return -1; // not found
4 }
5 void bubble_sort(int *a, int n){
6     for(int i=0; i<n-1; i++){
7         for(int j=0; j<n-1-i; j++){
8             if(a[j]>a[j+1]){ int t=a[j]; a[j]=a[j+1]; a[j+1]=t; }
9         }
10    }

```

`linear_search` is $O(n)$; `bubble_sort` is $O(n^2)$. The sorted array can then be binary-searched in $O(\log n)$ — a theme of Ch. 7's `std::sort/find`.

3.6 Nested Structs and Typedef

Structs can contain other structs, enabling hierarchical records.

```

1 struct Date{ int day, month, year; };
2 struct Student{
3     int id;

```

```

4     char name[32];
5     float gpa;
6     struct Date enrolled;
7 };
8 typedef struct Student Student; // alias: Student instead of struct Student
9 Student s = {123, "Asha", 4.8f, {15, 8, 2024}};
10 printf("%02d/%02d/%04d\n", s.enrolled.day, s.enrolled.month, s.enrolled.year);
11 Student *ps = &s;
12 ps->gpa = 4.9f; // -> for pointer

```

`typedef` shortens declarations; modern C++ prefers using `Student = struct Student;`. Nested access chains like `s.enrolled.year` read left to right: “student’s enrolment date’s year”.

3.7 Common Pitfalls and Safe String Handling

C’s string functions are notoriously unsafe. Prefer bounded variants and always ensure termination.

```

1 char dst[8];
2 // UNSAFE: strcpy(dst, "Hello, NTU!"); // 12 bytes into 8 -- overflow!
3 snprintf(dst, sizeof(dst), "%s", "Hello, NTU!"); // safe: truncates to "Hello, "
4 printf("dst='%s' len=%zu\n", dst, strlen(dst)); // dst='Hello, ' len=7
5
6 char a[16]="Hello", b[]=" World";
7 // strncat with remaining space
8 strncat(a, b, sizeof(a)-strlen(a)-1); // leaves room for \0
9 printf("%s\n", a); // Hello World
10
11 // Tokenising (destructive)
12 char line[]="one,two,three";
13 for(char *tok=strtok(line,","); tok; tok=strtok(NULL,",")) printf("[%s]\n", tok);

```

Rule: every `char` buffer has a size; every string op must respect it. When in doubt, use `snprintf` and check its return value (needed length).

3.8 Chapter Notes

C strings are a frequent source of buffer overflows; modern code prefers bounded functions (`snprintf`, `strncpy`). C++ `std::string` (Ch. 7) eliminates the terminator bookkeeping entirely.

3.9 Exercises

E3.1 **sizeof trap.** `int a[10];` What are `sizeof(a)` and `sizeof(&a)`? Write a function `void foo(int a[10])` and explain why `sizeof(a)` inside `foo` differs.

E3.2 **String length.** Implement `int my_strlen(const char *s)` without using `<string.h>`. Why must the parameter be `const char*`?

- E3.3 **Off-by-one.** This copies unsafely: `char dst[8]; strcpy(dst, "Hello, NTU!");` How many bytes are written? Fix it with `snprintf` and explain the truncation behaviour.
- E3.4 **Struct padding.** Given `struct S{char c; int i; char d;};` predict `sizeof(struct S)` on a 4-byte-aligned machine. Verify with `printf("%zu", sizeof(struct S))`.
- E3.5 **Record design.** Define `struct Course{char code[8]; int au; float grade;}` and write `float gpa(struct Course *cs, int n)` computing the AU-weighted GPA.

Chapter 4

Dynamic Memory: The Heap and Its Discipline

“Memory management is the price of power.” Stack memory is automatic but fixed-size; heap memory is manual but flexible. This chapter teaches you to wield `malloc` and `free` safely.

From zero: stack vs. heap as two regions. The stack grows/shrinks automatically per call; the heap is a pool you request chunks from. We draw both, then build `malloc/free` discipline with `valgrind` as proof.

Learning Outcomes

After this chapter you will be able to:

- (L1) contrast stack and heap lifetime and size;
- (L2) allocate/free heap memory with `malloc/calloc/realloc/free`;
- (L3) diagnose leaks, double-free, and use-after-free with `valgrind/ASan`;
- (L4) implement a resizable array (vector) in C;
- (L5) state and apply the “every `malloc` has a `free`” discipline.

4.1 Stack vs. Heap

```
1 void foo(void){
2     int local = 5;           // stack: freed when foo returns
3     int *heap = malloc(sizeof(int)); // heap: lives until free(heap)
4     if(!heap) return;       // malloc can fail!
5     *heap = 5;
6     free(heap); heap = NULL; // release + null to avoid dangling
7 }
```

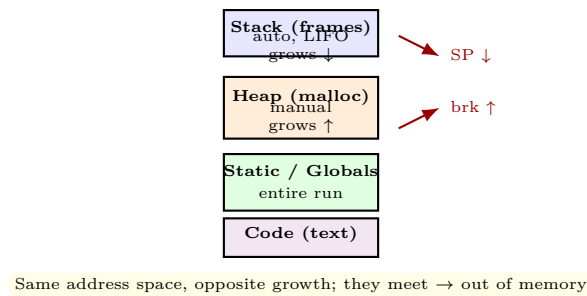


Figure 4.1: Process address space (simplified): stack grows down automatically; heap grows up on demand. Stack is per-call, heap is per-programmer-request.

Stack allocation is instant and reclaimed on return; heap allocation is explicit and survives the creating function — but you must free it.

4.2 malloc, calloc, realloc, free

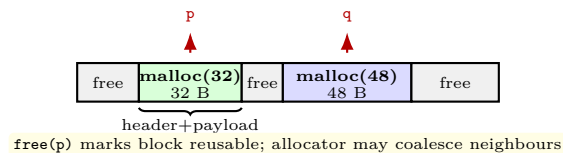


Figure 4.2: Heap as a sequence of blocks. `malloc` finds a free chunk, marks it allocated, returns its payload address. `free` marks it free again.

```

1  #include <stdlib.h>
2  #include <string.h>
3  int *a = malloc(10 * sizeof(*a));           // 10 ints, uninitialised
4  int *b = calloc(10, sizeof(*b));           // 10 ints, zeroed
5  if(!a || !b){ perror("malloc"); exit(1); }
6
7  a[0]=42; b[0]=42;                          // use
8  int *tmp = realloc(a, 20*sizeof(*a));      // grow to 20
9  if(tmp){ a=tmp; } else { /* handle failure, old a still valid */ }
10
11 free(a); free(b); a=b=NULL;                // always free what you malloc

```

Rules: check for `NULL`; `realloc` may move the block (use `tmp`); `free(NULL)` is safe no-op; never `free` stack addresses or the same pointer twice.

`sizeof(*a)` is preferred over `sizeof(int)` — it stays correct if `a`'s type changes.

4.3 Common Heap Bugs

- **Leak:** `malloc` without `free` — memory grows until exhaustion. Long-running programs crash.
- **Double free:** `free(p); free(p);` — corrupts allocator metadata.
- **Use-after-free:** `free(p); *p=1;` — writes to reclaimed memory; may corrupt next allocation.

- **Buffer overflow:** writing past the allocated size — corrupts adjacent block headers.

4.4 Valgrind and AddressSanitizer

Tools turn silent UB into loud errors. Compile with `-g` (debug symbols), then:

```

1 // leaky.c
2 int main(void){ malloc(100); return 0; } // leak!

gcc -g -Wall leaky.c -o leaky
valgrind --leak-check=full ./leaky          # reports 100 bytes lost
gcc -g -fsanitize=address leaky.c -o leaky_asan && ./leaky_asan # ASan report

```

Valgrind output names the allocation site and bytes lost; ASan aborts on the first illegal access with a stack trace. Run one of them on every heap-using program during development. On macOS, use `leaks` or ASan (valgrind is Linux-first).

4.5 A Resizable Array in C

We combine structs + heap to build a minimal vector:

```

1 typedef struct { int *data; int size; int cap; } Vec;
2 void vec_init(Vec *v){ v->data=NULL; v->size=0; v->cap=0; }
3 void vec_push(Vec *v, int x){
4     if(v->size==v->cap){
5         int nc = v->cap ? v->cap*2 : 4;
6         int *nd = realloc(v->data, nc*sizeof(*nd));
7         if(!nd){ perror("realloc"); exit(1); }
8         v->data=nd; v->cap=nc;
9     }
10    v->data[v->size++]=x;
11 }
12 void vec_free(Vec *v){ free(v->data); v->data=NULL; v->size=v->cap=0; }

```

Doubling amortises `realloc` cost to $O(1)$ per push. Every `vec_init` must be paired with `vec_free`.



Figure 4.3: Resizable array: heap buffer with spare capacity. Doubling keeps amortised $O(1)$ push and wastes at most half the buffer.

4.6 Deep Copy vs. Shallow Copy

When a struct owns heap memory, copying the struct copies the pointer, not the buffer — a shallow copy. Two structs then point to the same buffer; freeing both double-frees.

```

1 typedef struct { char *name; int age; } Person;
2 Person a; a.name = malloc(16); strcpy(a.name, "Asha"); a.age=20;
3 Person b = a; // shallow: b.name == a.name (same pointer!)
4 free(a.name); // b.name now dangling -- use-after-free if accessed
5 // Fix: deep copy
6 Person c; c.name = malloc(strlen(a.name)+1); strcpy(c.name, a.name); c.age=a.age;

```

In C++ (Ch. 5,8) copy constructors automate deep copying. In C you must do it manually, every time.

4.7 Worked Example: Dynamic String Builder

We build a string of unknown length by doubling, mirroring Vec but for chars.

```

1 #include <string.h>
2 typedef struct { char *buf; int len; int cap; } StrBuilder;
3 void sb_init(StrBuilder *sb){ sb->buf=malloc(16); sb->cap=16; sb->len=0; sb->buf[0]='\0';
4     }
5 void sb_append(StrBuilder *sb, const char *s){
6     int slen=strlen(s);
7     while(sb->len+slen+1 > sb->cap){
8         sb->cap*=2;
9         char *nb=realloc(sb->buf, sb->cap);
10        if(!nb){ perror("realloc"); exit(1); }
11        sb->buf=nb;
12    }
13    strcpy(sb->buf+sb->len, s); sb->len+=slen;
14 }
15 void sb_free(StrBuilder *sb){ free(sb->buf); sb->buf=NULL; sb->len=sb->cap=0; }

```

Every `sb_init` pairs with `sb_free`. The doubling keeps total copy cost $O(n)$ for n appended characters.

4.8 Chapter Notes

Heap allocators (first-fit, best-fit, segregated lists) are covered in SC2005 Operating Systems. For C++ alternatives that automate freeing, see Ch. 8 (RAII, smart pointers).

4.9 Exercises

E4.1 **Lifetime.** Why does returning `&local` from a function fail, but returning `malloc`'d memory succeeds? When must the caller `free` the latter?

E4.2 **calloc vs malloc.** When is `calloc(n, sizeof(T))` preferable to `malloc(n*sizeof(T))`? What does “zeroed” mean for pointer members?

E4.3 **realloc pitfall.** Explain why `a = realloc(a, new_size)` leaks on failure. Write the safe pattern with a temporary pointer.

- E4.4 **Valgrind reading.** Write a program that leaks 3 blocks and double-frees one. Run under valgrind/ASan and explain each diagnostic line.
- E4.5 **Vec shrink.** Extend `Vec` with `vec_pop` and a shrink policy (e.g. halve when `size == cap/4`). Argue why shrinking at `size == cap/2` could cause thrashing.

Chapter 5

C++ Classes and Object-Oriented Basics

“C++ is C with a better abstraction toolkit.” A `class` bundles data with the operations that maintain its invariants — turning a loose struct into a disciplined object.

From zero: C structs $\rightarrow$ C++ classes. We start with why bare structs are error-prone (anyone can break invariants), then add constructors, encapsulation, and methods step by step. No prior OOP assumed.

Learning Outcomes

After this chapter you will be able to:

- (L1) define a C++ class with access control (`private/public`);
- (L2) write constructors, destructors, and use member-initialiser lists;
- (L3) distinguish value vs. reference semantics and use `const` correctly;
- (L4) explain encapsulation and class invariants;
- (L5) trace object lifetime (construction $\rightarrow$ use $\rightarrow$ destruction).

5.1 From Struct to Class

A C struct is passive: any code can set `s.gpa = 999`. A class makes data `private` and exposes only safe operations.

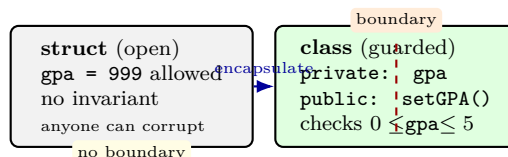


Figure 5.1: Encapsulation: a class erects a wall. Outside code must use the public interface, which can enforce invariants.

```

1 #include <string>
2 class Student{
3 private:
4     int id_;
5     std::string name_;
6     double gpa_;           // invariant: 0.0 <= gpa_ <= 5.0
7 public:
8     Student(int id, std::string name, double gpa) // constructor
9         : id_(id), name_(std::move(name)), gpa_(gpa) {
10         if(gpa_<0 || gpa_>5) throw std::out_of_range("gpa");
11     }
12     double gpa() const { return gpa_; }           // getter, const = won't mutate
13     void setGPA(double g){                        // setter with check
14         if(g<0||g>5) throw std::out_of_range("gpa");
15         gpa_=g;
16     }
17     std::string name() const { return name_; }
18 };

```

class defaults to private; struct defaults to public — otherwise identical in C++.

5.2 Constructors, Destructors, Initialiser Lists

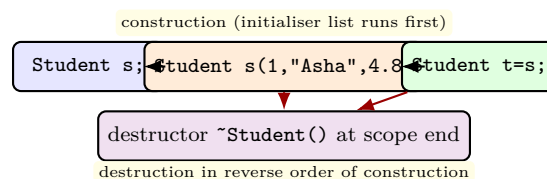


Figure 5.2: Object lifetime: construction via initialiser list, then use, then automatic destruction when the scope ends. Copy construction is discussed in Ch. 8.

```

1 class Vec2{
2     double x_, y_;
3 public:
4     Vec2(): x_(0), y_(0) {}           // default constructor
5     Vec2(double x, double y): x_(x), y_(y) {} // parameterised
6     // member initialiser list : x_(x) runs BEFORE body -- required for const/ref members
7     double norm() const { return std::sqrt(x_*x_+y_*y_); }
8     ~Vec2(){ /* release resources if any */ } // destructor
9 };
10 Vec2 a;           // default
11 Vec2 b(3,4);      // param
12 Vec2 c = b;       // copy (memberwise for now; Ch.8 refines)

```

Prefer initialiser lists over assignment in the body: they directly construct members, while assignment constructs then overwrites.

5.3 Encapsulation and Invariants

A class invariant is a predicate that every public operation must preserve (e.g. $0 \leq \text{gpa} \leq 5$, vector `size` $\leq$ `capacity`). By making data private and checking in setters/constructors, the invariant holds for all clients. Think of the class as a tiny machine with a valid-state set; public methods are the only buttons.

```

1  class BoundedCounter{
2      int val_, max_;
3  public:
4      BoundedCounter(int m): val_(0), max_(m) {}
5      void inc(){ if(val_ < max_) ++val_; }
6      void reset(){ val_=0; }
7      int get() const { return val_; }
8      // invariant: 0 <= val_ <= max_ (always, between calls)
9  };

```

5.4 Const Correctness and References

```

1  void print(const Student& s){           // const ref: no copy, no mutate
2      std::cout << s.name() << " " << s.gpa() << "\n";
3  }
4  Student s(1, "Asha", 4.8);
5  print(s);                             // binds without copying string
6  // s.gpa() is const member function -- callable on const Student&

```

`const` on a method means `this` is `const`; it can be called on `const` objects. Pass large objects by `const&` to avoid copies. Return by value for small types, by `const` ref for members (with lifetime care).

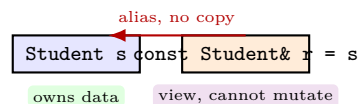


Figure 5.3: Reference semantics: `const Student&` is an alias (no copy) with read-only access. Value semantics would copy the whole object.

5.5 Operator Overloading

Classes can give meaning to operators like `+`, `==`, `<<` so objects feel like built-in types.

```

1  class Vec2{
2      double x_, y_;
3  public:
4      Vec2(double x=0, double y=0): x_(x), y_(y) {}
5      Vec2 operator+(const Vec2& o) const { return {x_+o.x_, y_+o.y_}; }
6      bool operator==(const Vec2& o) const { return x_==o.x_ && y_==o.y_; }
7      double x() const { return x_; }
8      double y() const { return y_; }
9      friend std::ostream& operator<<(std::ostream& os, const Vec2& v){
10         return os<<" ("<<v.x_<<" "<<v.y_<<") ";

```

```

11     }
12 };
13 Vec2 a(1,2), b(3,4);
14 Vec2 c = a+b;           // (4,6)
15 std::cout<<c<<"\n";    // (4,6)

```

Overload only when semantics are obvious; `+` should not surprise. Stream insertion `operator<<` is idiomatically a `friend` non-member.

5.6 Static Members and Friends

`static` members belong to the class, not to any instance; `friend` grants selective access to private data.

```

1 class Counter{
2     static int total_;    // shared across all objects
3     int id_;
4 public:
5     Counter(): id_(++total_) {}
6     static int total(){ return total_; }
7     friend void debug(const Counter& c); // can read c.id_ private
8 };
9 int Counter::total_ = 0; // define static outside class
10 void debug(const Counter& c){ std::cout<<c.id_<<"\n"; }

```

Static data needs an out-of-class definition (or `inline static` since C++17). Friends break encapsulation minimally — prefer member functions when possible.

5.7 Member Functions vs. Non-Members

Not every operation should be a member. Prefer non-member non-friend when it can be implemented via the public interface — it reduces coupling and keeps the class minimal.

```

1 class Vec2{
2     double x_, y_;
3 public:
4     Vec2(double x=0, double y=0): x_(x), y_(y) {}
5     double x() const { return x_; }
6     double y() const { return y_; }
7     Vec2& operator+=(const Vec2& o){ x_+=o.x_; y_+=o.y_; return *this; }
8 };
9 // non-member built from public members + +=
10 Vec2 operator+(Vec2 a, const Vec2& b){ a+=b; return a; }
11 double dot(const Vec2& a, const Vec2& b){ return a.x()*b.x()+a.y()*b.y(); }
12 bool isZero(const Vec2& v){ return v.x()==0 && v.y()==0; }

```

If a function needs private access, make it a member or friend; otherwise keep it outside. This is the “minimal interface” principle (Meyers).

5.8 Chapter Notes

Stroustrup *The C++ Programming Language* (4th ed.) and Meyers *Effective C++* are the classic guides. Rule of Zero/Three/Five (when to write copy/move/destructor) is covered in Ch. 8.

5.9 Exercises

- E5.1 **Invariant.** Design `class Date{int d,m,y}` with invariant “valid calendar date”. Which operations must validate? Sketch constructor and `setDay`.
- E5.2 **Initialiser list.** Why must a `const int id_` or `int& ref_` member be initialised in the member-initialiser list, not in the body?
- E5.3 **Const.** Mark which methods should be `const` in a `BankAccount` class (`balance()`, `deposit()`, `withdraw()`, `getId()`). What breaks if you omit `const`?
- E5.4 **Encapsulation break.** Show how returning a non-const reference to a private member (`int& getVal()`) breaks encapsulation. Fix it.
- E5.5 **Lifetime.** Trace construction/destruction order for `{ Student a(1,"A",4); Student b=a; }` How many constructor/destructor calls? Which kind?

Chapter 6

Inheritance, Polymorphism, and Templates

“Inheritance is sharing; polymorphism is substituting; templates are stamping.” Three ways to reuse code, each with a distinct cost model.

From zero: reuse without copy-paste. We start with why duplication hurts, then derive is-a hierarchies, virtual dispatch, and generic stamping from first principles.

Learning Outcomes

After this chapter you will be able to:

- (L1) model is-a relationships with public inheritance;
- (L2) explain virtual dispatch and the vtable mechanism;
- (L3) use templates for generic functions and classes;
- (L4) distinguish static (compile-time) vs. dynamic (run-time) polymorphism;
- (L5) apply access specifiers and virtual destructors correctly.

6.1 Inheritance: Is-A

If `Dog` is-a `Animal`, every `Dog` has the `Animal` part plus its own extensions. Inheritance captures this.

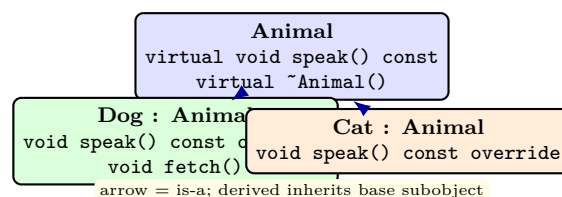


Figure 6.1: Inheritance hierarchy: `Dog` and `Cat` each contain an `Animal` subobject. Public inheritance models is-a.

```
1 struct Animal{
```

```

2   virtual void speak() const { std::cout<<"...\n"; }
3   virtual ~Animal() = default;           // virtual dtor: essential!
4 };
5 struct Dog : public Animal{
6     void speak() const override { std::cout<<"Woof!\n"; }
7     void fetch() const { std::cout<<"*fetches*\n"; }
8 };
9 Dog d; Animal& a = d; // Dog is-an Animal: binds base ref to derived
10 a.speak();           // Woof! (virtual dispatch)

```

`override` makes the compiler check the signature matches; always use it. Make base destructors `virtual` so `delete` through base pointer destroys the derived part.

6.2 Polymorphism and Virtual Dispatch

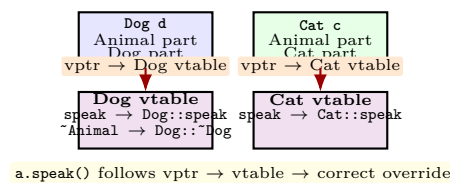


Figure 6.2: Virtual dispatch: each object holds a hidden `vptra` to its class's vtable. A virtual call indirections through it, selecting the derived override at run time.

```

1 void chorus(const std::vector<Animal*>& v){
2     for(auto *a: v) a->speak(); // dynamic dispatch: Dog woofs, Cat meows
3 }
4 Dog d; Cat c;
5 chorus({&d,&c}); // Woof! Meow!

```

Cost: one extra indirection per virtual call. Non-virtual calls are direct (zero overhead). Prefer non-virtual by default; make virtual only when you need run-time substitution. Pure virtual (`=0`) makes a base abstract: `virtual void draw() const =0;`

6.3 Templates: Compile-Time Stamping

Templates generate code per type at compile time (static polymorphism). No vtable, no indirection — but one instantiation per distinct type.

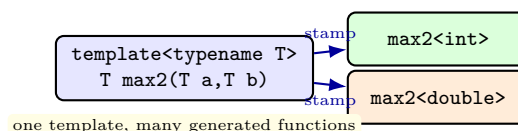


Figure 6.3: Templates as stamps: the compiler generates a separate function/class per `T`. No run-time cost, but code bloat if overused.

```

1 template<typename T>
2 T max2(T a, T b){ return (a>b)?a:b; }
3 int x = max2<int>(3,7);           // explicit

```

```

4 double y = max2(3.1, 2.9);           // deduced: T=double
5 // Class template
6 template<typename T>
7 struct VecT{ T x,y; T dot(const VecT& o) const { return x*o.x + y*o.y; } };
8 VecT<double> v{1.0,2.0};

```

Templates are type-checked after instantiation; errors can be verbose. C++20 concepts constrain T with readable diagnostics (preview in Ch. 8).

6.4 Static vs. Dynamic Polymorphism

	Templates (static)	Virtual (dynamic)
When chosen	compile time	run time
Cost	code size per T	one vtable indirection
Heterogeneous container	no (one T per container)	yes (<code>vector<Animal*></code>)

Rule of thumb: use templates for generic algorithms on uniform types; use virtual for heterogeneous runtime substitution.

6.5 Access Control and Protected Members

public inheritance keeps is-a; private/protected inheritance is implementation reuse, not is-a. protected members are visible to derived classes but not to outsiders.

```

1 class Base{
2 protected: int prot_ = 42; // derived can see, outsiders cannot
3 private:   int priv_ = 0;
4 public:    int pub_ = 1;
5 };
6 class Derived : public Base{
7 public:
8     void foo(){ prot_ = 7; /* ok */ /* priv_=1; error */ }
9 };
10 Base b; // b.prot_ error -- protected not accessible outside

```

Prefer private data with protected accessors over protected data: it preserves the base's invariant while still allowing customisation.

6.6 Abstract Classes and Interfaces

A class with at least one pure virtual function (=0) is abstract — it cannot be instantiated, only derived from. This models interfaces.

```

1 struct Drawable{ // interface: pure virtual
2     virtual void draw() const = 0;

```

```

3     virtual ~Drawable() = default;
4 };
5 struct Circle : Drawable{
6     double r; explicit Circle(double r_): r(r_) {}
7     void draw() const override { std::cout<<"Circle r="<<r<<"\n"; }
8 };
9 struct Square : Drawable{
10    double s; explicit Square(double s_): s(s_) {}
11    void draw() const override { std::cout<<"Square s="<<s<<"\n"; }
12 };
13 void render(const std::vector<Drawable*>& v){
14     for(auto *d: v) d->draw(); // uniform interface
15 }

```

Interfaces decouple callers from concrete types — the foundation of plugin architectures and test mocks. In modern C++, pure abstract classes are preferred over deep hierarchies.

6.7 Template Specialisation and Non-Type Parameters

Templates can be specialised for specific types or take compile-time constants as parameters.

```

1 template<typename T, int N> // non-type param: array size known at compile time
2 struct Array{
3     T data[N];
4     T& operator[](int i){ return data[i]; }
5     constexpr int size() const { return N; }
6 };
7 Array<int,5> a; // 5 ints, no heap, size() is compile-time
8
9 // Function template with overload for const char*
10 template<typename T> void print2(T x){ std::cout<<x<<"\n"; }
11 template<> void print2<const char*>(const char* s){ std::cout<<"C-string: " <<s<<"\n"; }
12 print2(42); // int version
13 print2("hello"); // specialised for const char*

```

Non-type parameters enable fixed-size containers without dynamic allocation; specialisation lets you tailor behaviour per type while keeping a uniform call site.

6.8 Chapter Notes

Sutter & Alexandrescu *C++ Coding Standards* Ch. 37–40 covers inheritance discipline. STL (Ch. 7) is almost entirely templates; inheritance appears sparingly there by design.

6.9 Exercises

- E6.1 **Slicing.** `Animal a = Dog(); a.speak();` What prints and why? How does `Animal& a = d;` differ? Explain object slicing.
- E6.2 **Virtual dtor.** Give a program where deleting a `Dog` through `Animal*` without a virtual destructor leaks resources. Fix it.
- E6.3 **Template deduction.** For `template<typename T> void f(T a, T b)`, what happens with `f(1, 2.0)`? How to fix the call or the template?
- E6.4 **Override check.** Remove `override` and misspell `speak` as `speal` in `Dog`. What happens with and without `override`? Why is `override` best practice?
- E6.5 **Design.** Model `Shape` with `area()` and `draw()`. Should they be virtual/pure virtual? Should `Shape` be instantiable? Justify and sketch the hierarchy.

Chapter 7

The Standard Template Library: Vectors, Maps, and Iterators

“STL is not a library you call — it is a language you think in.” Containers hold data; iterators traverse it; algorithms operate on it. Learn the trinity and you read modern C++ fluently.

From zero: we generalise Ch. 4’s Vec and Ch. 3’s arrays. STL containers automate the heap discipline; iterators generalise pointers; algorithms generalise loops. Pictures before code throughout.

Learning Outcomes

After this chapter you will be able to:

- (L1) use `vector`, `string`, `map/unordered_map`, `set` idiomatically;
- (L2) traverse any container with iterators and `range-for`;
- (L3) apply STL algorithms (`sort`, `find`, `accumulate`);
- (L4) state complexity guarantees of key containers;
- (L5) choose the right container for a task.

7.1 Sequence Containers: `vector` and `string`

`std::vector<T>` is Ch. 4’s resizable array done right: heap buffer, size, capacity, automatic growth, RAII cleanup.

```
1 #include <vector>
2 #include <string>
3 std::vector<int> v = {10,20,30};
4 v.push_back(40);           // may reallocate; amortised O(1)
5 v[0]=99; v.at(1)=21;       // at() bounds-checks, [] does not
6 for(int x: v) std::cout<<x<<" "; // range-for (sugar for iterators)
```

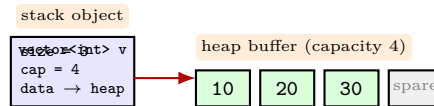


Figure 7.1: vector layout: small stack handle (size/cap/pointer) owns a heap array. Growth doubles capacity; spare slots avoid reallocation on push.

```

7  std::string s="Hello";
8  s += " NTU";           // string is vector<char> with extras
9  s.substr(0,5);         // "Hello"

```

Reserve when you know the size: `v.reserve(1000)` avoids repeated reallocations. References/pointers into a vector may dangle after reallocation — a common bug.

7.2 Associative Containers: map, set, unordered_map

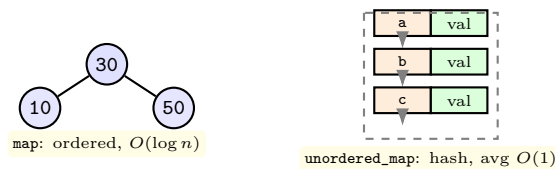


Figure 7.2: Ordered map (balanced BST, sorted) vs. unordered_map (hash table, unsorted but faster average).

```

1  #include <map>
2  #include <unordered_map>
3  #include <set>
4  std::map<std::string,int> scores;           // ordered by key
5  scores["Asha"]=95; scores["Bo"]=82;
6  for(auto &kv: scores) std::cout<<kv.first<<":"<<kv.second<<"\n"; // sorted
7
8  std::unordered_map<std::string,int> h = {"Asha",95};
9  h["Bo"]=82;                               // avg O(1)
10 if(h.find("Asha")!=h.end()) std::cout<<"found\n";
11 std::set<int> s={3,1,4,1,5};               // {1,3,4,5} sorted unique

```

Complexity: map/set $O(\log n)$ per op (red-black tree); unordered_map/unordered_set average $O(1)$, worst $O(n)$ on hash collision. Prefer unordered when order is irrelevant.

7.3 Iterators: Generalised Pointers

An iterator is a pointer-like object that knows how to step through a container. All STL algorithms speak iterators, not containers.

```

1  #include <algorithm>
2  #include <numeric>
3  std::vector<int> v={4,1,3,2};
4  std::sort(v.begin(), v.end());           // 1 2 3 4
5  auto it = std::find(v.begin(), v.end(), 3); // iterator to 3
6  if(it!=v.end()) std::cout<<"at " <<(it-v.begin())<<"\n";

```

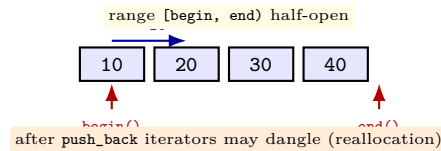


Figure 7.3: Iterator range `[begin, end)`: `begin` points to first element, `end` is one past the last. Increment advances; dereference reads.

```

7  int sum = std::accumulate(v.begin(), v.end(), 0); // 10
8  // range-for is sugar:
9  // for(auto it=v.begin(); it!=v.end(); ++it) ...
10 for(int &x: v) x*=2; // mutate via ref

```

Iterator categories: random-access (vector), bidirectional (map), forward (forward_list). Algorithms state which they require.

7.4 Choosing a Container

Need	Use	Why
Resizable array	<code>vector</code>	cache-friendly, $O(1)$ push
Sorted lookup	<code>map/set</code>	$O(\log n)$, ordered
Fast lookup, no order	<code>unordered_map</code>	avg $O(1)$
Unique sorted values	<code>set</code>	dedup + order

7.5 STL Algorithms in Depth

Algorithms are function templates that operate on iterator ranges. They never modify container structure directly; they read/write through iterators.

```

1  #include <algorithm>
2  #include <numeric>
3  std::vector<int> v={5,2,8,1,9};
4  // sort, reverse, count, transform
5  std::sort(v.begin(), v.end()); // 1 2 5 8 9
6  std::reverse(v.begin(), v.end()); // 9 8 5 2 1
7  int cnt = std::count(v.begin(), v.end(), 5); // 1
8  std::vector<int> w(v.size());
9  std::transform(v.begin(), v.end(), w.begin(),
10                [](int x){ return x*2; }); // w = 18 16 10 4 2
11 int sum = std::accumulate(v.begin(), v.end(), 0); // 25
12 bool any = std::any_of(v.begin(), v.end(),
13                        [](int x){ return x>8; }); // true

```

All algorithms work on any container with the required iterator category. Prefer algorithms over hand-written loops: they are optimised, self-documenting, and less error-prone.

7.6 Complexity Guarantees and Big-O of Containers

Operation	<code>vector</code>	<code>map</code>	<code>unordered_map</code>
<code>push_back</code>	amort. $O(1)$	—	—
<code>find</code>	$O(n)$	$O(\log n)$	avg $O(1)$
<code>insert</code>	$O(n)$ (middle)	$O(\log n)$	avg $O(1)$
<code>operator[]</code>	$O(1)$	$O(\log n)$	avg $O(1)$
Memory	contiguous	tree nodes	hash buckets

These guarantees drive container choice. `vector` wins for iteration (cache locality); `map` wins when order matters; `unordered_map` wins for raw lookup speed.

7.7 Chapter Notes

Josuttis *The C++ Standard Library* is the STL handbook. Complexity details are in the standard and at cppreference.com. Do not store raw owning pointers in containers; use values or smart pointers (Ch. 8).

7.8 Exercises

- E7.1 **Invalidation.** `auto it=v.begin(); v.push_back(99); *it=1;` When can this be UB? How does `reserve` help? What about `std::list`?
- E7.2 **Map vs unordered.** You need to count word frequencies and later print them alphabetically. Which map type? What if you only need fast lookup?
- E7.3 **Algorithm.** Use `std::sort + std::find` to sort a vector and locate a value. What iterator does `find` return on failure? How to check?
- E7.4 **Complexity.** State big- O for `vector::push_back`, `map::find`, `unordered_map::find` (average/worst). When does unordered degrade?
- E7.5 **Range-for desugar.** Rewrite `for(auto &x: v) x++` using explicit iterators. Explain why `auto &` vs. `auto` matters.

Chapter 8

Modern C++: RAII, Smart Pointers, and Move Semantics

“Modern C++ is not C with extras — it is a different language that happens to compile C.” RAII makes resource safety automatic; smart pointers make ownership explicit; move semantics makes copies cheap.

From zero: Ch. 4’s discipline was manual; now we automate it. Every resource gets an owner object whose destructor frees it. We build RAII, then smart pointers, then moves — each fixing a failure mode of the last.

Learning Outcomes

After this chapter you will be able to:

- (L1) explain RAII and why it prevents leaks even on exceptions;
- (L2) use `unique_ptr`, `shared_ptr`, `weak_ptr` correctly;
- (L3) distinguish lvalues from rvalues and apply move semantics;
- (L4) state the Rule of Zero/Three/Five;
- (L5) write exception-safe code with RAII types.

8.1 RAII: Resource Acquisition Is Initialisation

Idea: tie every resource (memory, file, lock) to an object’s lifetime. Constructor acquires; destructor releases. No manual `free/close` — the scope does it, even if an exception is thrown.

```
1 #include <fstream>
2 void process(){
3     std::ofstream out("log.txt"); // acquires file handle
```

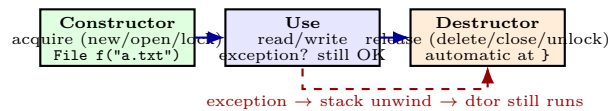


Figure 8.1: RAII timeline: acquisition in constructor, automatic release in destructor. Stack unwinding guarantees the destructor runs even on exception.

```
4     out << "hello\n";
5     if(some_error()) throw std::runtime_error("oops");
6     // no close needed -- destructor closes even on throw
7 } // out.~ofstream() runs here
```

`std::string`, `std::vector`, `std::lock_guard`, `std::fstream` are all RAII types. Manual `new/delete` is the non-RAII alternative — avoid it.

8.2 Smart Pointers: Explicit Ownership

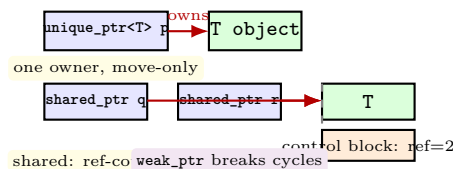


Figure 8.2: Smart pointers: `unique_ptr` has one owner; `shared_ptr` shares via a control block with reference count; `weak_ptr` observes without owning.

```

1 #include <memory>
2 auto p = std::make_unique<int>(42);           // unique ownership
3 // auto q = p;                               // error: not copyable
4 auto q = std::move(p);                       // transfer ownership; p==nullptr
5
6 auto a = std::make_shared<std::string>("hello"); // shared, ref count 1
7 auto b = a;                                   // ref count 2, same object
8 std::cout<<a.use_count()<<"\n";             // 2
9 std::weak_ptr<std::string> w = a;             // weak: doesn't increment count
10 // w.lock() -> shared_ptr if object still alive

```

Prefer `make_unique`/`make_shared` over raw `new`. Use `unique_ptr` by default; `shared_ptr` only when shared ownership is genuinely needed. `weak_ptr` breaks cycles (A owns B, B observes A weakly).

8.3 Move Semantics: Stealing Instead of Copying

Copying a vector duplicates its heap buffer ($O(n)$). *Moving* steals the buffer pointer and nulls the source ($O(1)$). Moves fire on *rvalues* (temporaries, `std::move`'d values).

```
1 std::vector<int> a = {1,2,3,4,5};
2 std::vector<int> b = a;           // copy: O(n)
3 std::vector<int> c = std::move(a); // move: O(1), a is now empty
4 // a is valid but unspecified -- only safe to assign/destroy
```

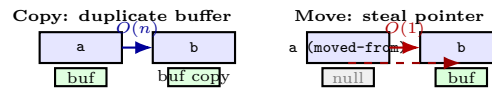


Figure 8.3: Copy duplicates the heap buffer; move transfers the pointer, leaving the source empty. Move is $O(1)$ vs. $O(n)$.

```
5 std::vector<int> make_vec(){ return {1,2,3}; } // NRVO/move elision
6 auto d = make_vec();                      // no copy (elided)
```

The compiler auto-moves temporaries; write `std::move` only for named lvalues you no longer need. After move, treat the source as empty.

8.4 Rule of Zero / Three / Five

- **Rule of Zero:** if your class uses RAII members (`string`, `vector`, smart pointers), write *no* custom destructor/copy/move — the compiler does it correctly.
- **Rule of Five:** if you *must* write one of destructor, copy-ctor, copy-assign, move-ctor, move-assign, you likely need all five.
- Prefer Zero; fall to Five only for low-level resource wrappers.

```
1 // Rule of Zero: no custom special members needed
2 class Student{
3     std::string name_; std::vector<int> scores_;
4 public:
5     Student(std::string n): name_(std::move(n)) {}
6     // compiler-generated ctor/copy/move are correct
7 };
```

8.5 Exception Safety and RAII in Practice

RAII makes exception safety almost free. Consider manual vs. RAII resource handling when an exception is thrown:

```
1 void manual(){
2     int *p = new int[100];
3     FILE *f = fopen("data.txt","r");
4     risky_operation(); // if throws, leak p and f!
5     delete[] p; fclose(f);
6 }
7 void raii(){
8     auto p = std::make_unique<int[]>(100);
9     std::ifstream f("data.txt"); // RAII file handle
10    risky_operation(); // if throws, destructors still run -- no leak
11 } // p and f destroyed automatically
```

Exception safety levels: *basic* (no leak), *strong* (rollback on failure), *nothrow* (never throws). RAI types give basic safety by default; transactions give strong safety.

8.6 Perfect Forwarding and Modern Idioms

C++11 introduced `auto`, range-for, lambdas, and forwarding references. C++17/20 added structured bindings and concepts.

```

1  auto add = [](auto a, auto b){ return a+b; }; // generic lambda (C++14)
2  std::cout<<add(3,4)<<" "<<add(3.1,2.9)<<"\n"; // 7 6.0
3
4  std::map<std::string,int> m={{ "Asha",95},{ "Bo",82}};
5  for(auto &[name,score]: m) // structured binding (C++17)
6      std::cout<<name<<": "<<score<<"\n";
7
8  // Perfect forwarding: preserve lvalue/rvalue nature
9  template<typename T>
10 void wrapper(T&& arg){ // forwarding reference
11     callee(std::forward<T>(arg)); // forwards as l- or r-value
12 }
```

These idioms reduce boilerplate and let the compiler generate optimal code. Prefer `auto` when the type is obvious; prefer range-for over index loops; prefer lambdas over hand-written functors.

8.7 Chapter Notes

Meyers *Effective Modern C++* (2014) is the definitive move/RAI guide. C++ Core Guidelines: “R.1: Manage resources automatically using RAI.”

8.8 Exercises

- E8.1 **RAI proof.** Write a function that opens two files and throws midway. Show RAI handles close it correctly while manual `fopen/fclose` leaks the second file.
- E8.2 **Ownership.** When would you choose `unique_ptr` vs. `shared_ptr` vs. raw `T*` (non-owning)? Give an example of each.
- E8.3 **Move trace.** `vector<int> a={1,2}; vector<int> b=std::move(a); cout<<a.size()<<" "<<b.size();` What prints? Is `a[0]` safe to read after the move?
- E8.4 **Rule of Zero.** A class holds `string name_` and `vector<double> grades_`. Do you need a custom destructor? What if it held `int* buf` from `new int[10]`?
- E8.5 **Cycle.** `struct A{shared_ptr<B> b;}; struct B{shared_ptr<A> a;};` Why does this leak? Fix it with `weak_ptr` and explain the ref counts.